

OMAR ABDOU'S CV

BASIC INFORMATION

- **Name:** Omar Abdou
- **Title:** Senior Software Engineer | Distributed Systems & Backend Architecture | AI Product Builder
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PROFESSIONAL SUMMARY

Over **9 years of experience** in the Software industry. I have designed and worked with fault-tolerant, scalable solutions in fintech, SaaS, AI, as well as other industries. A senior developer with a particular passion for AI and **distributed systems**. Applying concepts such as distributed consensus, consistency models, replication, and fault-tolerant system design from scratch in **my own distributed key-value database, Crystal DB**. As a co-founder and lead engineer at LifeOS Labs, I architected **Quik AI & Tafkeer AI**. 2 AI aggregator platforms integrating 44+ LLMs in the same system, alongside authentication, payment, and compliance requirements. Being written in Go and adhering strongly to optimization best practices, Quik AI's first response was **faster than OpenAI's ChatGPT** at a time. In 2016, I ran my own **software agency, Code Netro**, that **delivered over 12 projects in five different countries** for roughly 3 years until I joined a chain of corporate and startup companies from 2019 until today.

SKILLS

- **Programming Languages:** Go, Python, JavaScript, TypeScript, PHP
- **Distributed Systems:** State Machine Replication, Distributed Logs, Apache Kafka, Consensus Algorithms, Logical & Physical Clocks, Coordination Services, CAP Theorem, Consistency Models, Asynchronous Communication.
- **DevOps, Cloud & Infrastructure:** Docker, Kubernetes, Terraform, AWS, GitHub Actions
- **Backend & Systems Architecture:** REST/Restful API, gRPC, Event-Driven Architecture, Event Streaming & Messaging, Microservices, Domain-Driven Design (DDD), Go (Gin), Python (FastAPI, Django, Flask), Node.js (Express, NestJS), PHP
- **Databases:** Storage Engine Internals (B-Trees, LSM-Trees), ACID & Isolation Levels, Schema Encoding & Evolution, Replication & Sharding, PostgreSQL, Redis, MongoDB (NoSQL).
- **Frontend Development:** React, Next.js, Vue, Nuxt.js, and Vite

EXPERIENCE

Co-Founder & Lead Engineer

LifeOS Labs, Berlin, Jan 2025 – Present

- **Crystal DB:** A **distributed consensus-based** key-value database built from scratch in Go. Implements the **RAFT** consensus algorithm end-to-end. Including **leader election**, log **replication**, with a crash-safe **write-ahead log**, snapshot-based log **compaction** and a single-writer control loop driving per-peer replication goroutines. State machine is pluggable via a storage interface designed for **LSM-tree persistent storage**. Validated with unit tests and a real multi-node cluster integration suite **covering leader failure and data survival**.

- **Quik:** LifeOS's first AI aggregator and super-app, enabling users to chat, search, and research across **44+ LLMs** with full **vector- and graph-RAG memory**. As lead engineer I directed the complete lifecycle — from Figma design to deployment and DevOps. Built with Go, PostgreSQL, a **microservices backend**, and React.
- **Tafkeer AI:** AI aggregator with an autonomous routing layer selecting across 59 LLMs and specialized assistants — **the first of its kind in the region**. Extended the Quik platform architecture and adapted it for the Saudi market.

Independent Software R&D

Open-source, Remote / Berlin, Oct 2023 – Dec 2024

- **Netro:** An object-oriented **Python simulation** framework benchmarking hybrid Last-Mile Delivery (LMD) systems — a mix of autonomous delivery robots and trucks — against traditional methods. Built with **NumPy, Pandas, and Domain-Driven Design**.
- **Lugman:** An **LSTM recurrent-neural-network** trading assistant for automated market analysis and time-series forecasting, applying sequential deep-learning models to predictive decision-making. Built in **Python**.

Frontend developer

Kurnblick, Paderborn, part time, Oct 2021 – September 2023

- **Papenburg's Events management system:** A Rest API based system that streamlines the processes of event creation, handling customer data, ticketing, and digital payments. Allowing the organization to *manage more events with less* resources. Developed with **Vue.js, Nuxt.js, and Typescript**.

Backend developer

Simon Kucher, Bonn, Full time, Aug 2019 - Apr 2022

- **MacFit sales system:** An End-to-end sales system with a sophisticated recommendation engine and sales funnel tracking features of each custom across the multiple gym branches. I took the **initiative to step up as the project manager**. MacFit ended up *being the 3rd most profitable project throughout the company's international 40 offices* of the year. Developed with **Node.js, Vue.js, PostgreSQL, and Docker**
- **Precision Pricing:** A forecasting and optimization engine and an ETL designed for the Royal Bank of Canada. Developed with **Python**

Founder & Full stack Software developer

Code Netro, Online, Jan 2016 - July 2019

- **CodeNetro:** Started my own software development agency to keep up with the increased demand over my successful software development services and quality. During Code Netro more than **12 projects were delivered over 5+ countries**. Beneath are some honorable mentions. Mainly developer with **React, PHP, Laravel, NodeJs, VueJs, and MySQL**
- **Ridee** – Developed a Mobile app to arrange cycling communities & events in Berlin organized by Ridee GmbH. Developed with **Dart & Flutter**
- **Eonity's Pitch Master:** A music scheduler & trading platform based on Spotify's API and owned by the music label Eonity. Developed with **Javascript, Typescript, React, NodeJs, NestJs, and Mongo DB**
- **SunnyWorld:** Developed a commercial and an educational platform for the entrepreneurial company Sunny world. Transforming the company from a 2-people project into a **competing company with multiple offices** in the renewable energy sector. Developed with **PHP and MySQL**

EDUCATION

Masters of Data Science

International Business Administration Faculty, European University Viadrina Frankfurt (Oder) April 2022 – August 2025

- Data Science, Machine Learning, Data Visualization, Predictive Analytics, Statistical Modeling, Supply Chain, Forecasting, and SQL Databases

Bachelor of Business Administration

Business Faculty, Tanta University July 2016 – September 2021

- Business Administration, Organizational Behavior, Applied statistics, Project Management, Entrepreneurship, Business Digitalization, Database management

LANGUAGES

- English
- Arabic (Egypt)
- German

WOULD YOU LIKE TO SEND A MESSAGE?

Then feel free to reach out to my [LinkedIn](#) and I'd be happy to answer your questions!